Exercise 1

Set $f(x) = a_n x^n + \dots + a_1 x + a_0$ and $g(x) = n a_n x^{n-1} + \dots + a_1$

The first equation is equivalent to $f(x) = 0$ and the second one to $g(x) = 0$. Further we notice that $g(x) = f'(x)$.

For $k = 1, \dots, n-1$, we notice that $f(t_k) = f(t_{k+1}) = 0$ and f is continuous on $[t_k, t_{k+1}]$ and differentiable on (t_k, t_{k+1}) being a polynomian. Hence, for Rolle theorem $\exists\ s_k \in (t_k, t_{k+1})$ s.t. $f'(s_k) = g(s_k) = 0$.

Hence g has $n-1$ solutions $s_1, \dots, s_{n-1}$.

Exercise 2

a) We want to prove that $\lim_{x \to 0} \frac{f(x) - f(0)}{x}$ exists. This is equiv. to the existence of $\lim_{n \to \infty} \frac{f(x_n) - f(0)}{x_n}$ for any seq. $(x_n) \to 0$ with $x_n \neq 0$

Suppose $x_n > 0$. By hypotesis f is continuous on $[0, x_n]$ and differentiable on $(0, x_n)$. Hence $\exists\ t_n \in (0, x_n)$, by mean value thm s.t. $f(x_n) - f(0) = x_n \cdot f'(t_n)$. Since $t_n \in (0, x_n)$ $t_n \to 0$.

$$\lim_{n \to \infty} \frac{f(x_n) - f(0)}{x_n} = \lim_{n \to \infty} \frac{x_n f'(t_n)}{x_n} = \lim_{n \to \infty} f'(t_n) = f'(0)$$

where the last step holds by ~~that~~ existence ~~of~~ limit of f' at 0 given by hypotesis.

This equation proves the thesis.

b) It does not hold.

A contre example is $f(x) = x \sin(1/x)$

It is differentiable at $x \neq 0$ being composition and product of differentiable functions, and continuous at every $x \in \mathbb{R}$ from previous lectures.

If we pick $x_n = \frac{1}{\pi/2 + n\pi}$

$$\lim_{n \to \infty} \frac{f(x_n) - f(0)}{x_n} = \lim_{n \to \infty} \cancel{(\pi/2 + n\pi)} \frac{\sin(\pi/2 + n\pi)}{\cancel{\pi/2 + n\pi}}$$

$$= \lim_{n \to \infty} (-1)^{n+1}$$

which doesn't have a limit.

Exercise 3

a) f is continuous on $[-3,0]$ and on $(0,3]$.
We want to check continuity at 0

$$f(0) = \lim_{x\to 0^-} 6e^{-x^2} = 6e^0 = 6$$

$$\lim_{x\to 0^+} -x^3+6x^2+11x+6 = -0^3+6\cdot 0^2-11\cdot 0+6 = 6$$

They coincide and then f is continuous
Being continuous on the closed interval $[-3,3]$ f admits a maximum and a minimum by Weierstrass thm

b) We already know that f has a maximum and a minimum, from class we know that they can't be a point in the interior of $[-3,3]$ where the derivative exist and is non-zero.

$$f'(x) = \begin{cases} -2x\,e^{-x^2} & \text{for } x\in(-3,0) \\ -3x^2+12x-11 & \text{for } x\in(0,3) \end{cases}$$

It is not derivable at 0, since right and left limit of the derivative do not coincide there.
$-2xe^{-x^2}=0$ only at 0 but it doesn't belong to $(-3,0)$, so no point in $(-3,0)$ can be max or a min.
$-3x^2+12x-11=0$ at $x=2\pm\frac{1}{\sqrt{3}}$, they both belong to $(0,3)$ so all point, except these two, in $(0,3)$ can't be a max or a min.
The only points that can be max or min points for f are -3, 0, $2-1/\sqrt{3}$, $2+1/\sqrt{3}$ and 3. Let's compare the values that f takes at them.

$$f(-3) = e^{-9} < 1$$
$$f(0) = 6$$
$$f(2-1/\sqrt{3}) = \frac{2}{3\sqrt{3}} < 1$$
$$f(2+1/\sqrt{3}) = -\frac{2}{3\sqrt{3}}$$
$$f(3) = 0$$

Hence 0 is the maximum point for f
and $2+1/\sqrt{3}$ is the minimum point for f.

Exercise 4

a) Take $\alpha \in [0,1]$, ~~Thend~~ $x, y \in \mathbb{R}$.

Then $f+g(\alpha x + (1-\alpha)y) = f(\alpha x + (1-\alpha)y) + g(\alpha x + (1-\alpha)y)$

$$\leq \alpha f(x) + (1-\alpha)f(y) + \alpha g(x) + (1-\alpha)g(y)$$

$$= \alpha (f+g)(x) + (1-\alpha)(f+g)(y).$$

Hence $f+g$ is convex

b) Take $\alpha \in [0,1]$ and $x, y \in \mathbb{R}$

$g \circ f(\alpha x + (1-\alpha)y) \leq g(\alpha f(x) + (1-\alpha)f(y))$ by monotonicity of g and convexity of f

$\leq \alpha g(f(x)) + (1-\alpha) g(f(y))$

applying the def of convexity of g with $f(x)$ and $f(y)$.

c) Take $f(x) = (x-1)^2 + \varepsilon$, $g(x) = (x+1)^2 + \varepsilon$

They are convex because $f''(x) = g''(x) = 2 > 0$.

$fg(x) = (x-1)^2(x+1)^2 + \varepsilon[(x+1)^2 + (x-1)^2] + \varepsilon^2$

$fg(0) = 1 + 2\varepsilon + \varepsilon^2$

$fg(1) = fg(-1) = 4\varepsilon + \varepsilon^2$

For $\varepsilon = 1/4$ $fg(0) > fg(1) = fg(-1)$

This contradicts the def of convexity for $x = -1$, $y = 1$ and $\alpha = 1/2$.

Exercise 5

Let's write $x_i = e^{a_i}$, this is possible since $x_i > 0$ and we can set $a_i = \log(x_i)$.

The inequality in question becomes

$$\sqrt[n]{e^{a_1} \cdot \ldots \cdot e^{a_n}} \overset{?}{\leq} \frac{e^{a_1} \cdot \ldots \cdot e^{a_n}}{n}$$

$$\sqrt[n]{e^{a_1 + \ldots + a_n}} \overset{?}{\leq} \frac{e^{a_1} \cdot \ldots \cdot e^{a_n}}{n}$$

$$e^{\frac{a_1 + \ldots + a_n}{n}} \overset{?}{\leq} \frac{e^{a_1} + \ldots + e^{a_n}}{n}$$

This is equivalent to the Jensen inequality applied to e^x and $(a_1, \ldots, a_n)$ with weights $1/n$.

The only thing to check to apply it is the convexity of e^x.

But $(e^x)'' = e^x > 0$ which proves it.